Basic Emergency Care

Burn

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Learning objectives

- Define burn
- Identify causes of burn
- Identify burn depth and area of burn
- Know **primary survey (ABCDE)** in burn patients
- Learning initial emergency management

What is burn?

- A burn is an injury to the skin or other organic tissue primarily caused by heat or due to radiation, radioactivity, electricity, friction or contact with chemicals.

Causes of burn:

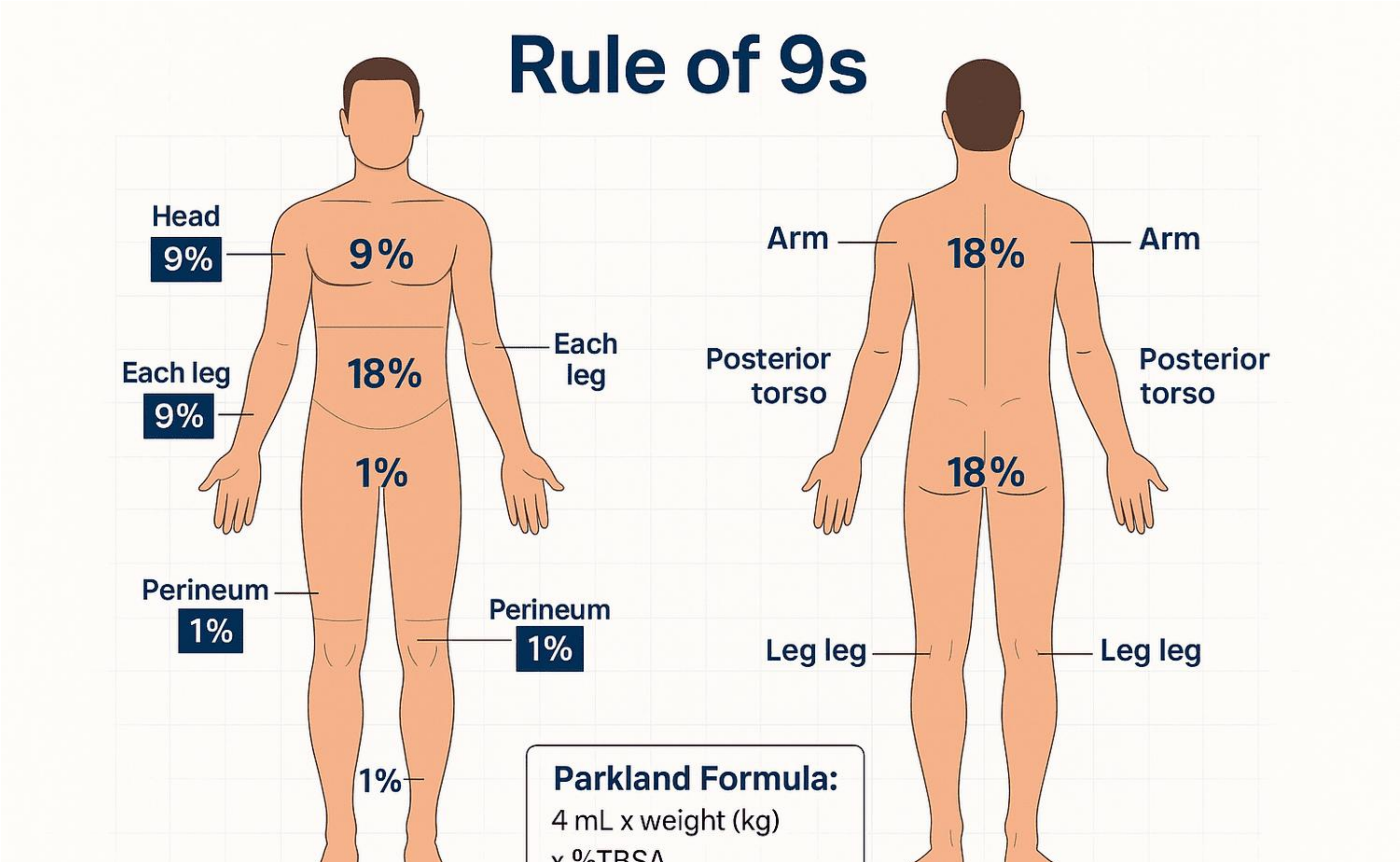
- 🔥 Thermal (Solid, Liquid, or Gas)
- ⚡ Electrical
- 🧪 Chemical
- ☢️ Radiation
- ☀️ Friction

Burn evaluation

- Estimation of the burn size (Surface area)
- Estimation of the burn depth (Thickness)

- Important determinants in treatment in prognosis

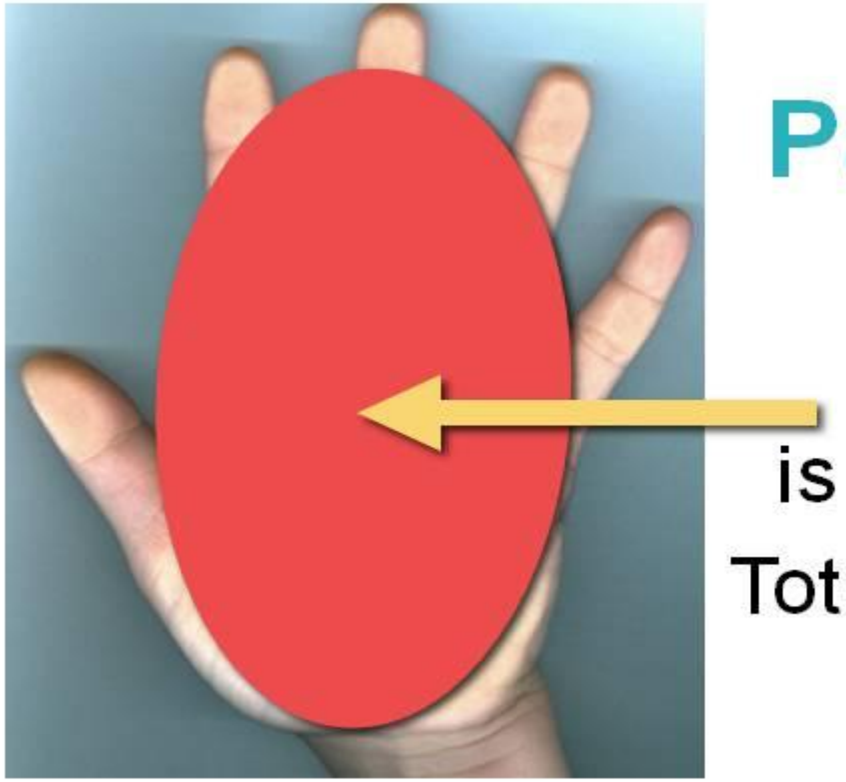
Burn size



Rule of 9s
Head
9%
9%
Each leg
9%
18%
Each
leg
1%
Perineum
1%
Perineum
1%
1%
Arm
18%
Arm
Posterior
torso
Posterior
torso
18%
Leg leg
Leg leg
Parkland Formula:
4 mL x weight (kg)

Burn size

Estimation of Small Burns



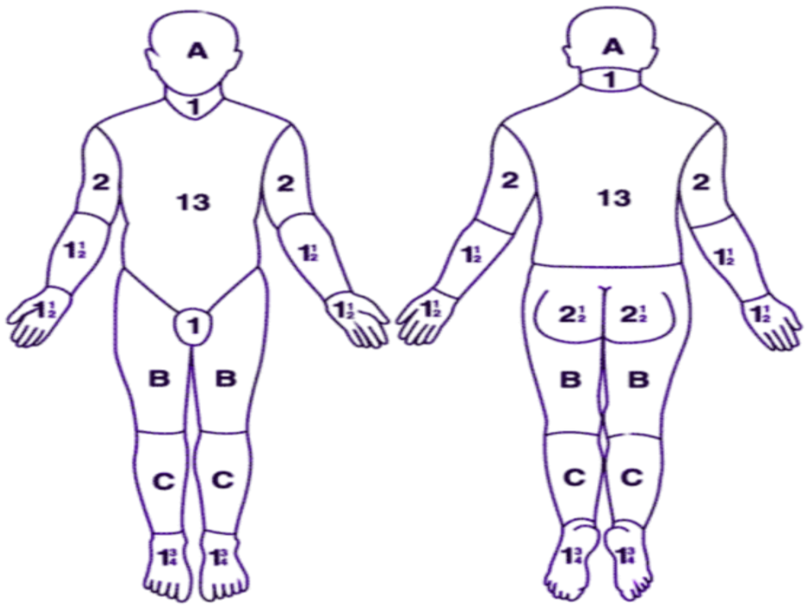
Palmar Method

Patient's palm
including fingers
is equal to 1% of their
Total Body Surface Area
(TBSA)

Burn size

Total Burn Surface Area

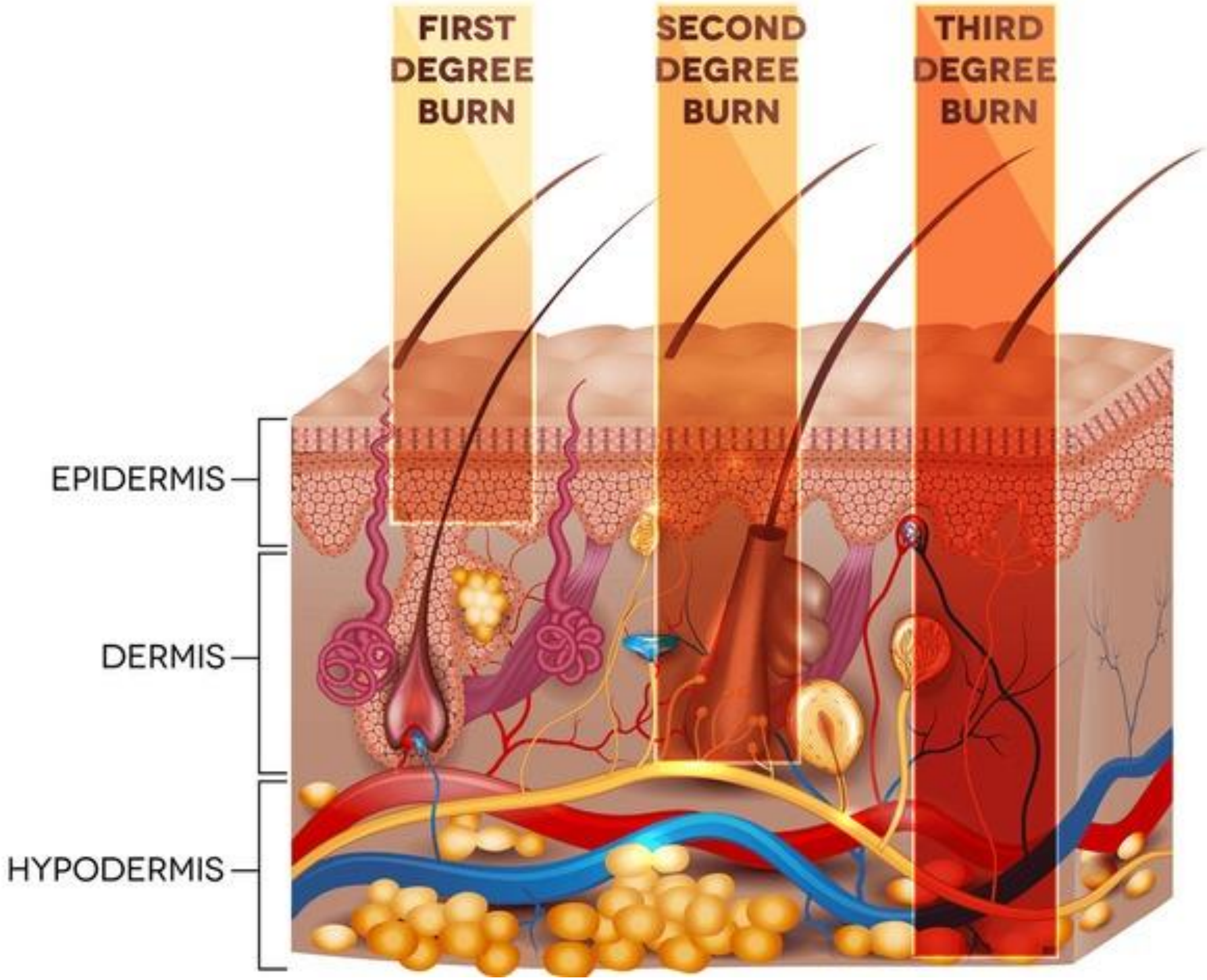
Lund and Browder Burn Chart



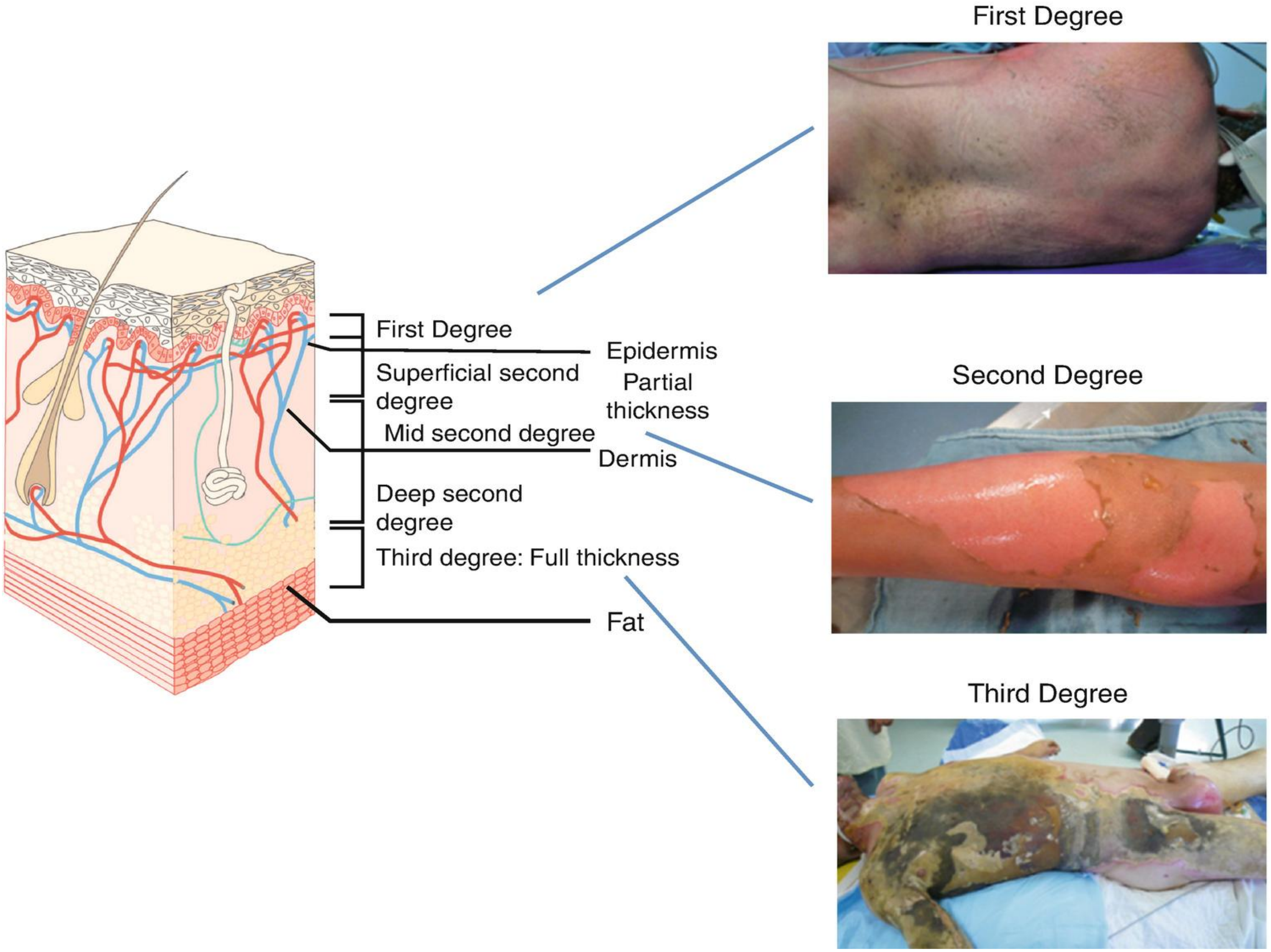
Age	0	1	5	10	15	Adult
A (Head)	9½%	8½%	6½%	5½%	4½%	3½%
B (Thigh)	2¾%	3¼%	4%	5%	6%	6½%
C (Leg)	2½%	2¾%	2¾%	3%	3¼%	3½%

Mark the areas of the burn on the body diagram.
Calculate the area using the age-based table below.
Do not include superficial burns in the calculation.

Burn depth


FIRST DEGREE BURN
SECOND DEGREE BURN
THIRD DEGREE BURN
EPIDERMIS
DERMIS
HYPODERMIS

Burn depth



First Degree
Second Degree
Third Degree
First Degree
Superficial second degree
Mid second degree
Deep second degree
Third degree: Full thickness
Epidermis
Partial thickness
Dermis
Fat

First degree burn

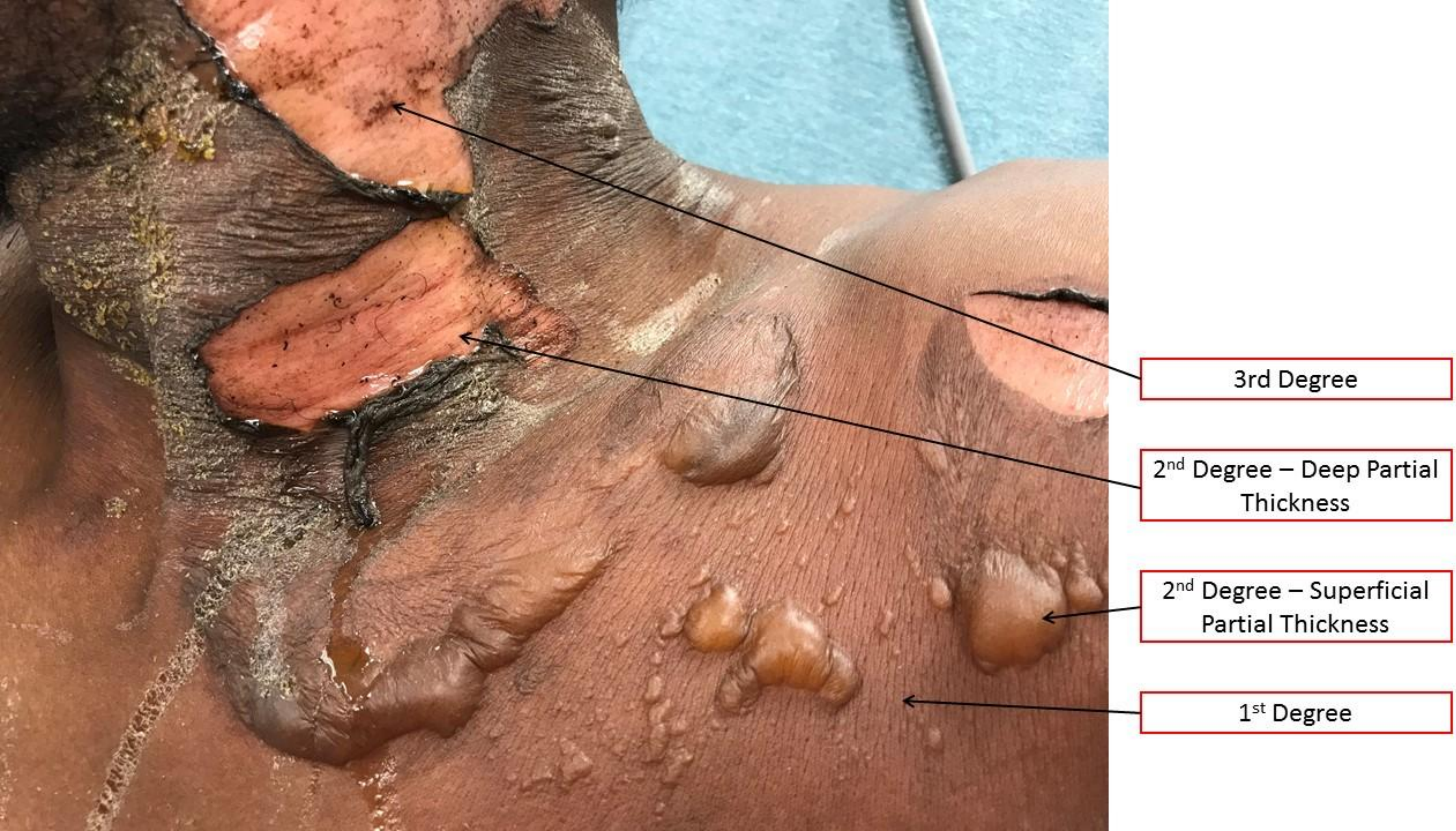
- Epidermis only
- Red, dry, painful
- No blisters
- Heals in 3–7 days
- No scar formation (Dermis is preserved)
- Example: sunburn

Second degree burn

- Epidermis + superficial part of dermis
- Pink, moist surface
- Blisters present (pathognomonic)
- Very painful
- Little or no scar formation
- Heals in 2–3 weeks

Third degree burn

- All skin layers destroyed
- White/brown/charred
- Dry, leathery (eschar)
- Painless (nerve destruction)
- Requires surgery (skin graft)


3rd Degree
2nd Degree – Deep Partial Thickness
2nd Degree – Superficial Partial Thickness
1st Degree

Management

- D: Danger
- R: Response
- A: Airway
- B: Breathing
- C: Circulation

Danger

- Do not attempt rescue if there is ongoing danger!
- Remove the person from the cause of the burn.
- If clothes are burning, make the person lie down to keep smoke away from their face.
- Use water, blanket or roll the person on the ground to smoother the flames.
- Once the burning has stopped, remove the clothing.
- Remove any wet clothing (Heat retention)
- flush the injured area with water “Irrigation” (In chemical burn only)

Airway

- Look for:
- Facial burns
- Singed nasal hair
- Soot in mouth
- Hoarseness of voice
- Extensive burns

- Give:
- High flow O2
- Attempt Endotracheal intubation early one if indicated and/or suspicion for impending airway compromise

Breathing

- High flow O2
- Ventilatory support
- Be aware of CO poisoning

Circulation

- Check pulse, BP
- Insert **2 large-bore IV lines immediately preferably through unburnt skin** (Through the burn is OK if needed)
- Control bleeding if present
- Start IV fluids immediately and calculate the dosage according to **Parkland's formula**

Treatment

- Assessment
- IV fluids, IV fluids, IV fluids
- Pain control
- Antibiotics
- Anti-tetanus prophylaxis
- Stress gastritis (curling ulcer) prophylaxis
- Irrigation (If chemical injury)
- Urinary catheter
- Nasogastric tube
- Blood investigations (Hb, PCV, RFT, electrolyte, arterial blood gas and carboxy Hb)

Complications

- Early:
- Shock
- Airway obstruction
- Electrolyte imbalance

- Late:
- Infection
- Contractures
- Hypertrophic scars
- Psychological trauma

Key takeaways

- ✅ Always follow DRABC
  ✅ Calculate area and depth
  ✅ Start fluids early
  ✅ Give oxygen
  ✅ Pain control
  ✅ Refer severe burns

Thank you